

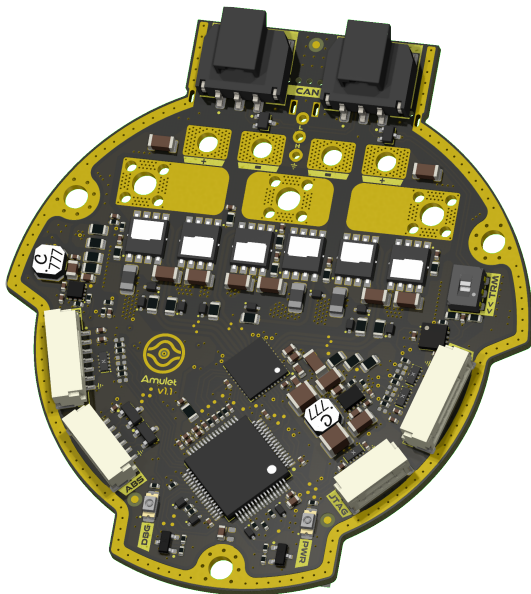
# Amulet Motion Controller

Variant: NO\_XT60

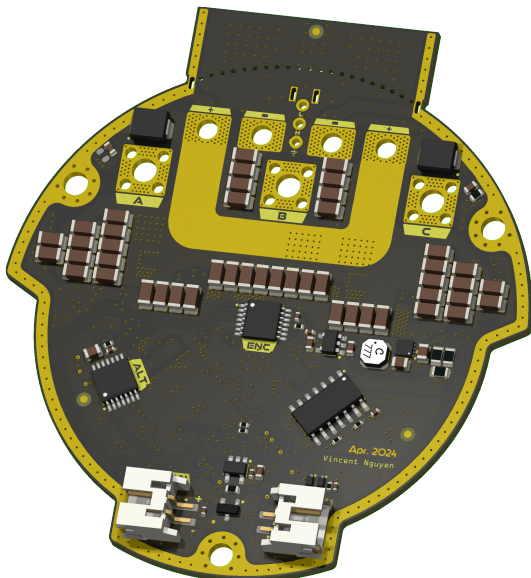
2024-11-19  
Rev 1.1

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## TOP VIEW



## BOTTOM VIEW



## DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

## NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

**DRAFT** - Very early stage of schematic, ignore details.

**PRELIMINARY** - Close to final schematic.

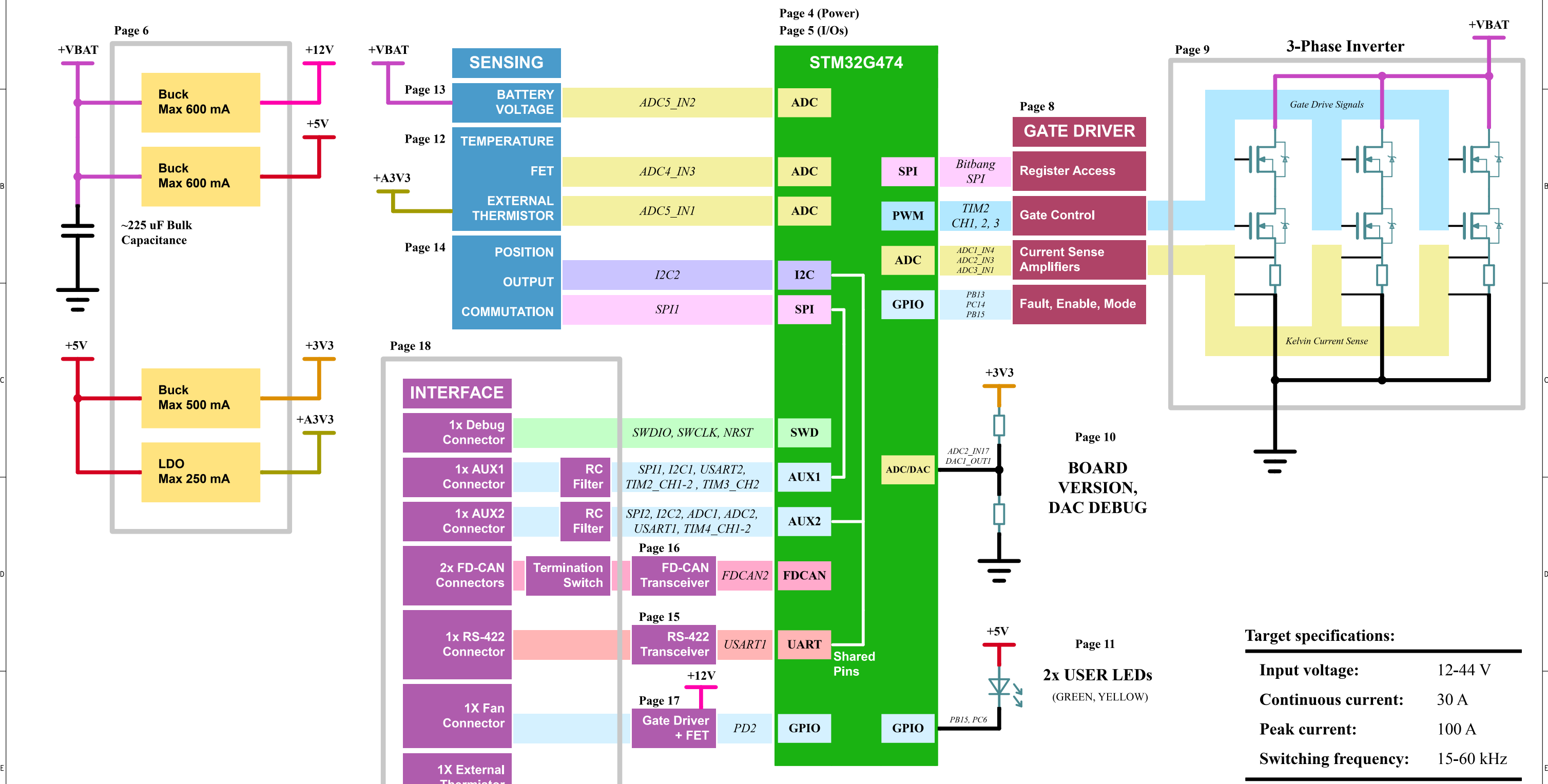
**CHECKED** - There shouldn't be any mistakes. Contact the engineer if you find any.

**RELEASED** - A board with this schematic has been sent to production.

Date: 13-APR-2024

|                  |                            |                                           |                             |                                                                                       |                             |                      |
|------------------|----------------------------|-------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|----------------------|
|                  | Comments:                  | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                  |                            | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                      |
|                  | Sheet Title:<br>Cover Page | File Name:<br>amulet_controller.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13                                                                   | Revision:<br>1.1            |                      |
| Sheet Path:<br>/ |                            |                                           | Reviewer:                   | Size:<br>A3                                                                           | Sheet:<br>1 of 21           |                      |

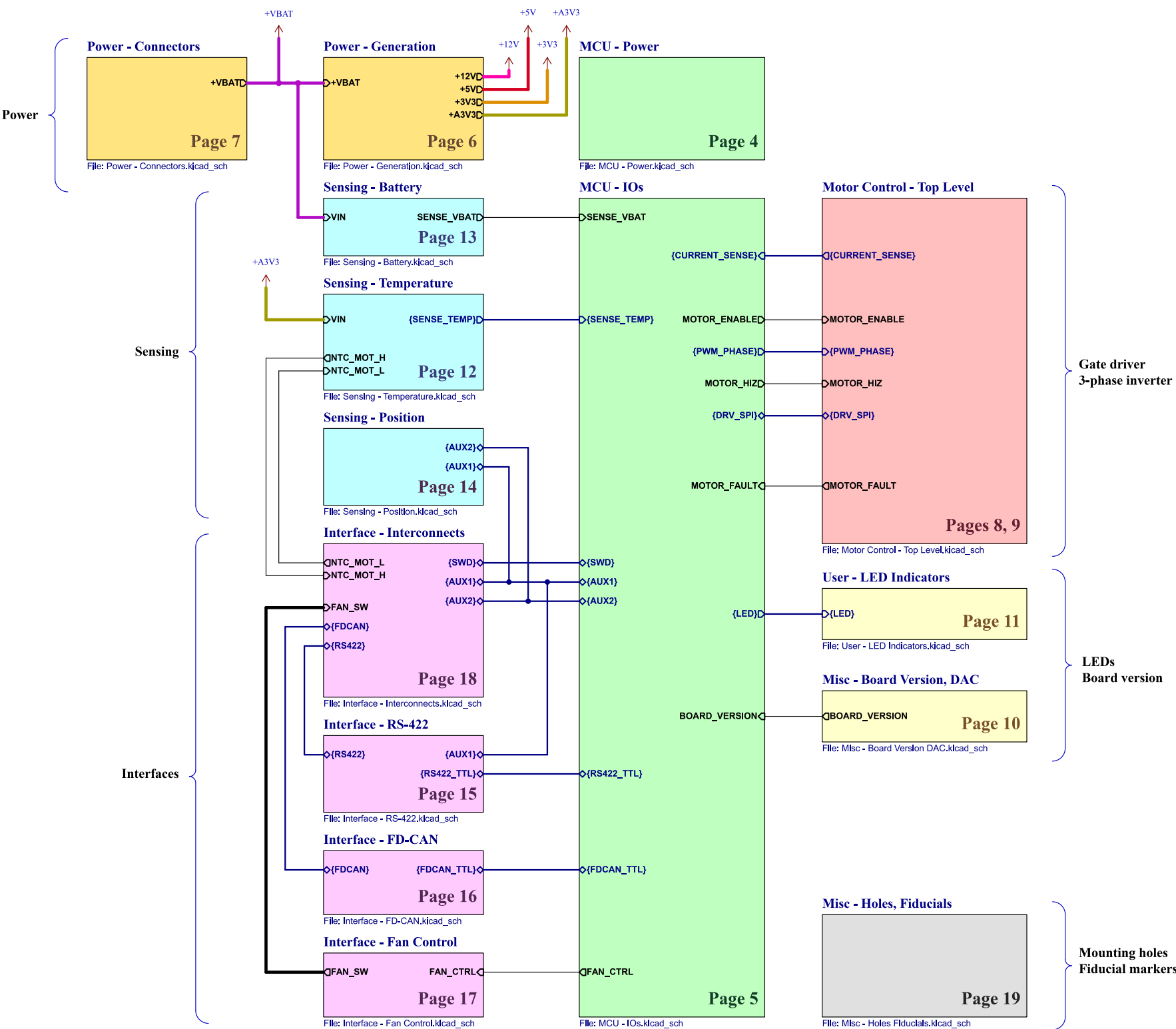
# [2] Block Diagram



| Target specifications: |           |
|------------------------|-----------|
| Input voltage:         | 12-44 V   |
| Continuous current:    | 30 A      |
| Peak current:          | 100 A     |
| Switching frequency:   | 15-60 kHz |

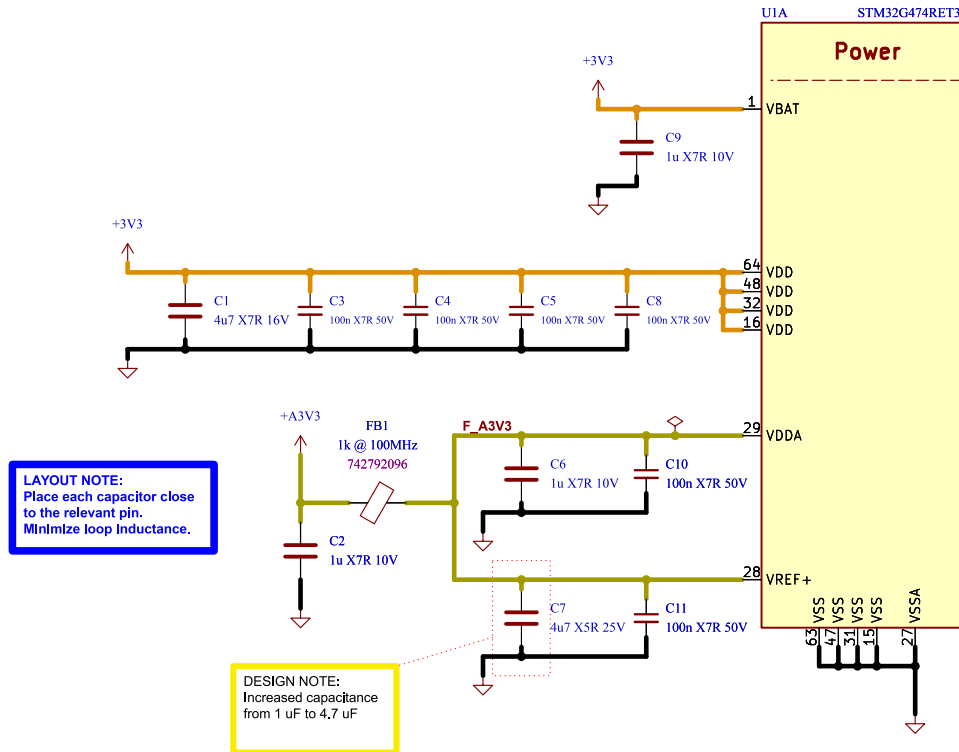
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|                                | Comments:                     | Company:<br>EPFL Xplore Research        |                             |  | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                                |                               | Board Name:<br>Amulet Motion Controller |                             |                                                                                       | Project Name:<br>Chienpanzé |                      |
|                                | Sheet Title:<br>Block Diagram | File Name:<br>Block Diagram.kicad_sch   | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13                                                                   | Revision:<br>1.1            |                      |
| Sheet Path:<br>/Block Diagram/ |                               | Reviewer:                               | Size:<br>A3                 | Sheet:<br>2 of 21                                                                     |                             |                      |

# [3] Project Architecture



|                                       |                                                |                             |                                                                                       |                                    |                      |
|---------------------------------------|------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|------------------------------------|----------------------|
| Comments:                             | Company:<br>EPFL Xplore Research               |                             |  | Variant:<br>NO_XT60                | Git Hash:<br>74ecd17 |
|                                       | Board Name:<br><b>Amulet Motion Controller</b> |                             |                                                                                       | Project Name:<br><b>Chienpanzé</b> |                      |
| Sheet Title:<br>Project Architecture  | File Name:<br>Project Architecture.kicad_sch   | Designer:<br>Vincent Nguyen | Date:<br>2023-12-22                                                                   | Revision:<br>1.1                   |                      |
| Sheet Path:<br>/Project Architecture/ |                                                | Reviewer:                   | Size:<br><b>A3</b>                                                                    | Sheet:<br><b>3</b> of <b>21</b>    |                      |

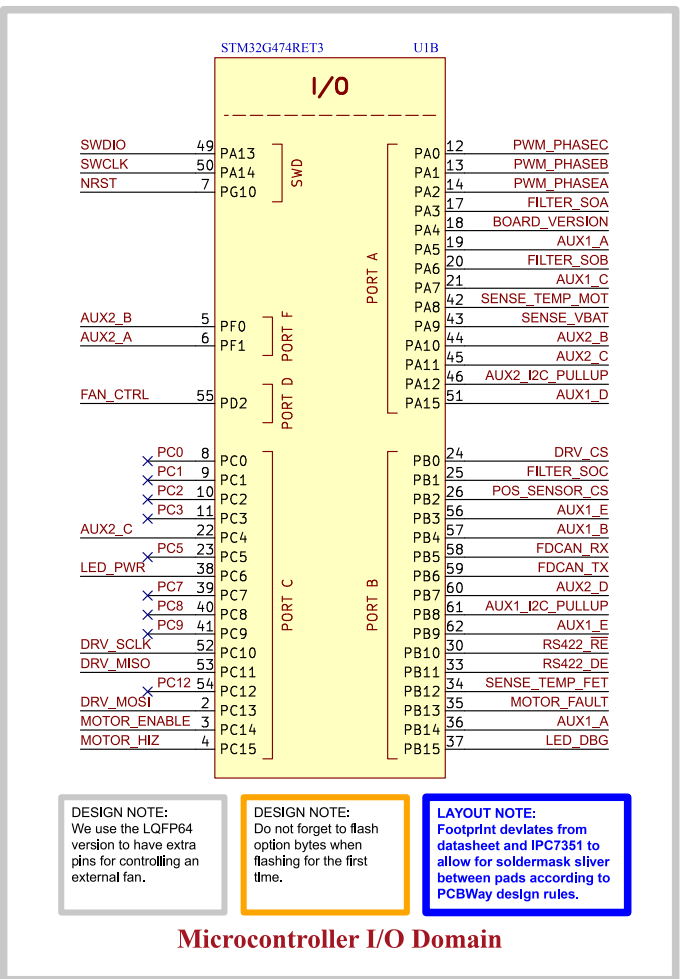
## [4] MCU - Power



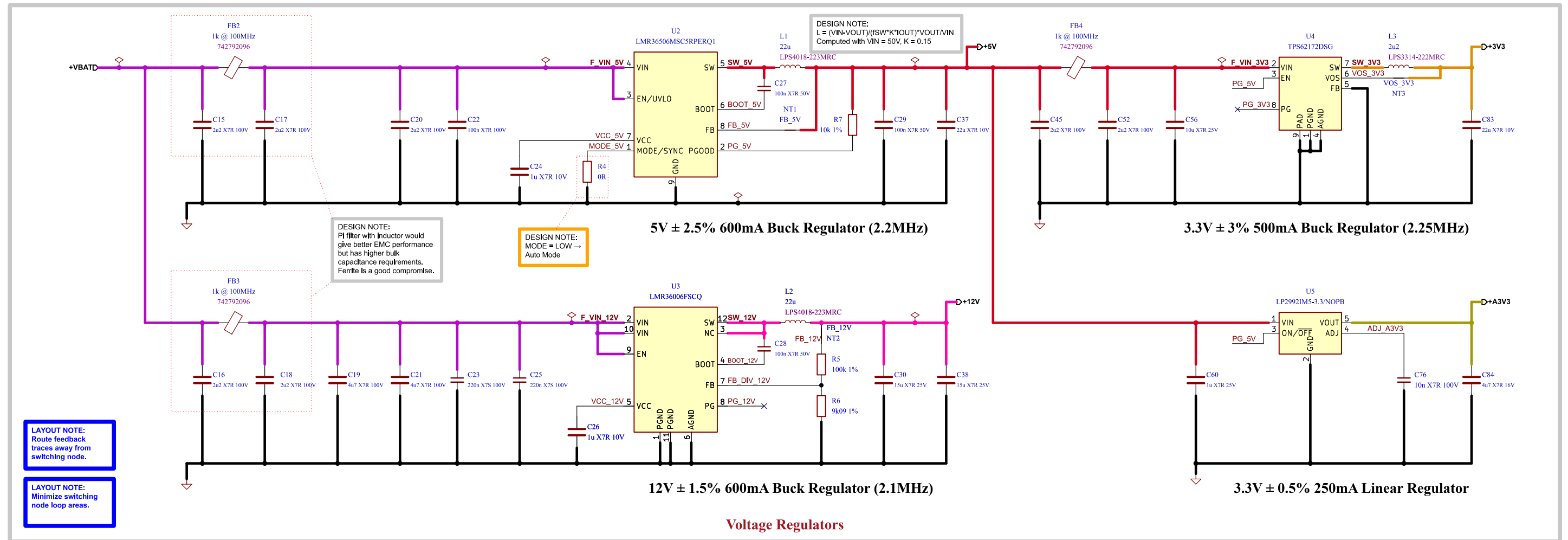
Microcontroller Power Supply Scheme

|  |                                                                                       |                                                       |                                                                                       |                                           |                                        |
|--|---------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------|
|  | <b>Comments:</b><br>AN5346<br>STM32G474 Datasheet p.81<br>J. Pieper ADC investigation | <b>Company:</b><br>EPFL Xplore Research               |  | <b>Variant:</b><br>NO_XT60                | <b>Git Hash:</b><br>74ecd17            |
|  |                                                                                       | <b>Board Name:</b><br><b>Amulet Motion Controller</b> |                                                                                       | <b>Project Name:</b><br><b>Chienpanzé</b> |                                        |
|  | <b>Sheet Title:</b><br>MCU - Power                                                    | <b>File Name:</b><br>MCU - Power.kicad_sch            | <b>Designer:</b><br>Vincent Nguyen                                                    | <b>Date:</b><br>2023-12-18                | <b>Revision:</b><br>1.1                |
|  | <b>Sheet Path:</b><br>/Project Architecture/MCU - Power/                              |                                                       | <b>Reviewer:</b>                                                                      | <b>Size:</b><br><b>A4</b>                 | <b>Sheet:</b><br><b>4</b> of <b>21</b> |

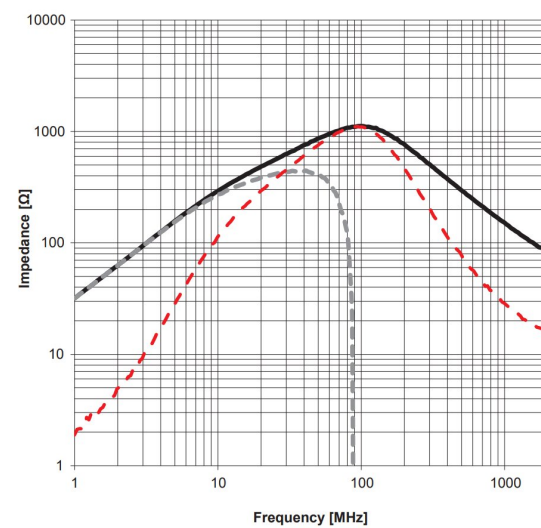
# [5] MCU - I/Os



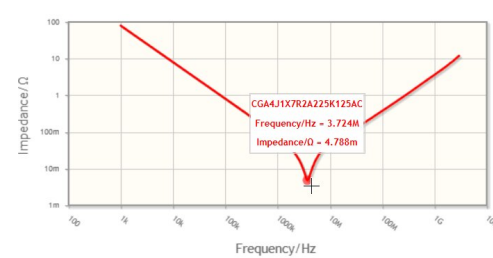
# [6] Power - Generation



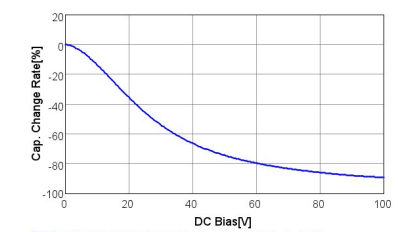
Ferrite bead Impedance vs Frequency curve



Pi filter capacitor Impedance vs Frequency curve



Bulk capacitance Derating curve



Capacitor and Ferrite Bead Parameters

Battery

+VBAT TP39 +VBAT  
+VBAT TP40 +VBAT  
+VBAT TP41 +VBAT

Filtered Power Rails

F\_VIN\_12V TP42 F\_VIN\_12V  
F\_VIN\_5V TP43 F\_VIN\_5V  
F\_VIN\_3V3 TP44 F\_VIN\_3V3

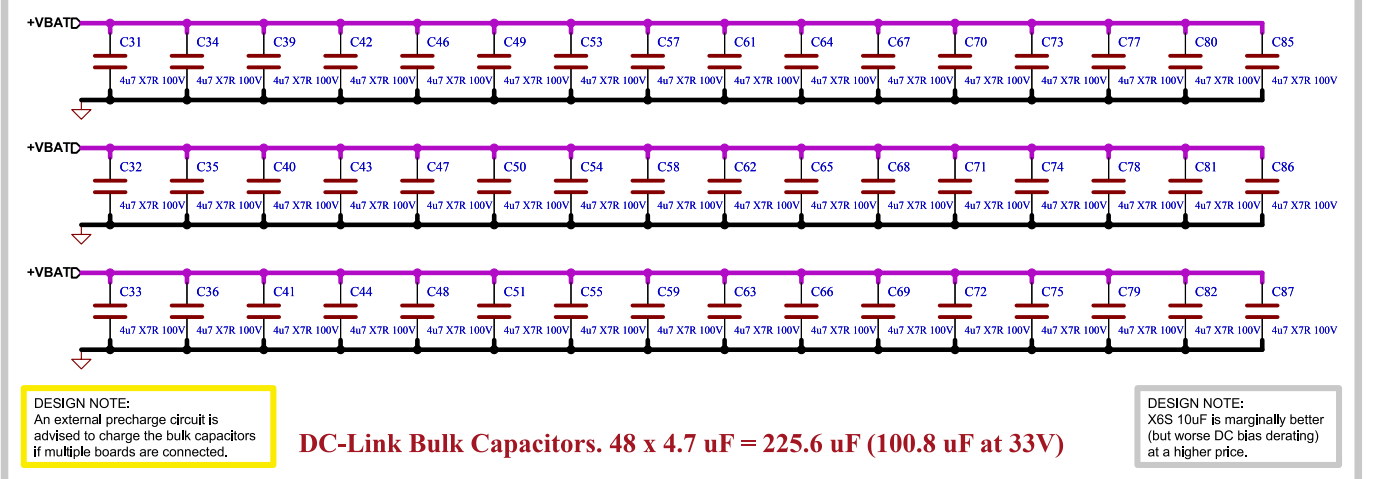
Power Rails

+12V TP45 +12V  
+5V TP46 +5V  
+5V TP47 +5V  
+3V3 TP48 +3V3  
+3V3 TP49 +3V3  
+3V3 TP50 +3V3  
+A3V3 TP51 +A3V3

Ground

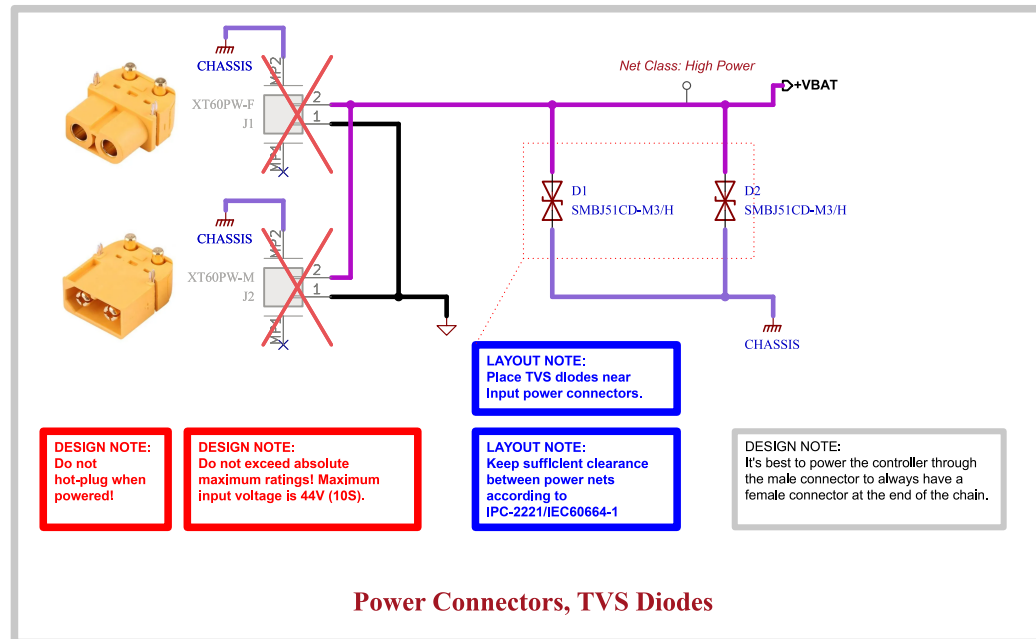
GND TP52 GND  
GND TP53 GND  
GND TP54 GND  
GND TP55 GND  
GND TP56 GND  
GND TP57 GND  
GND TP58 GND

Test Points



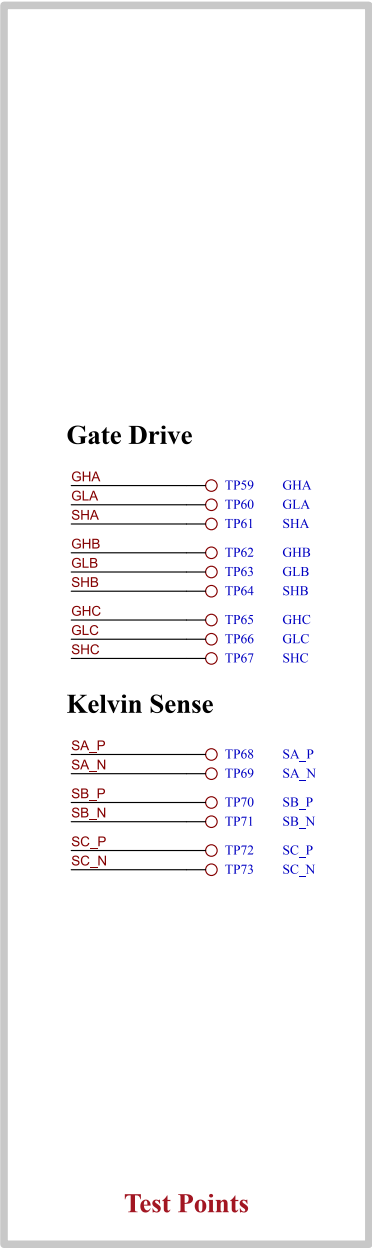
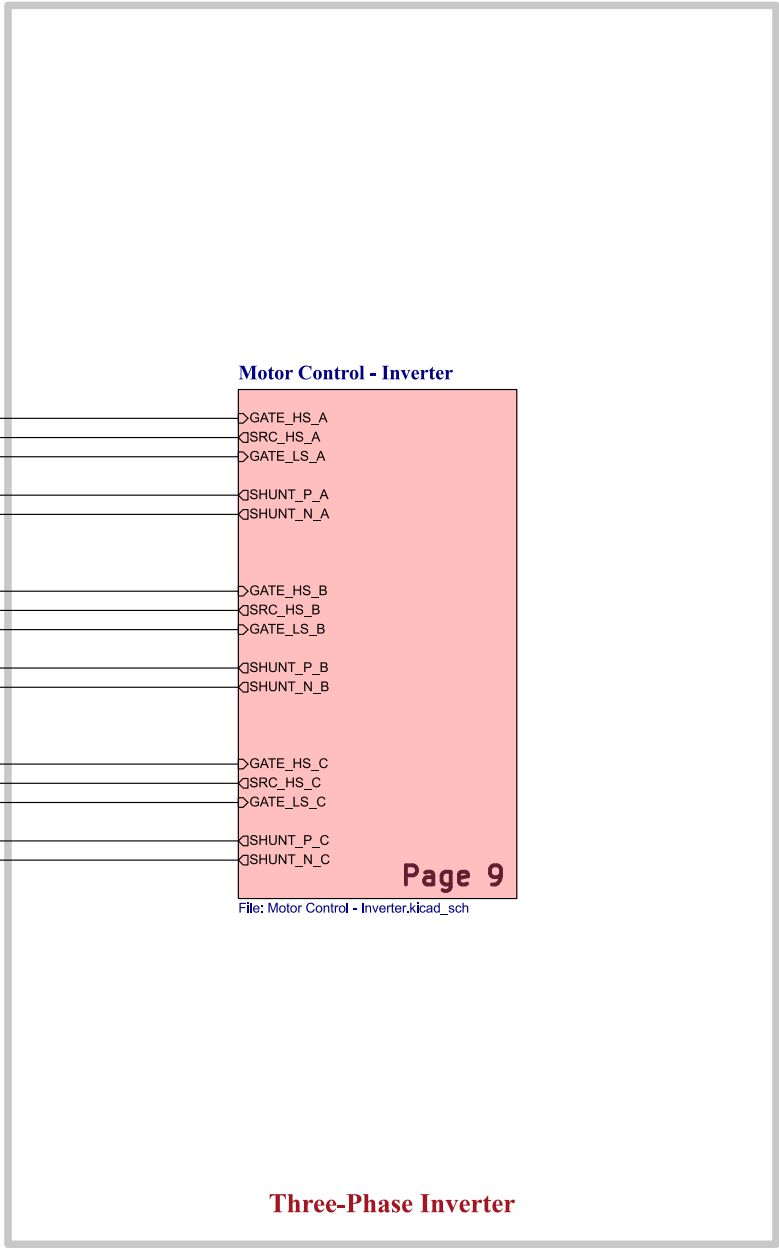
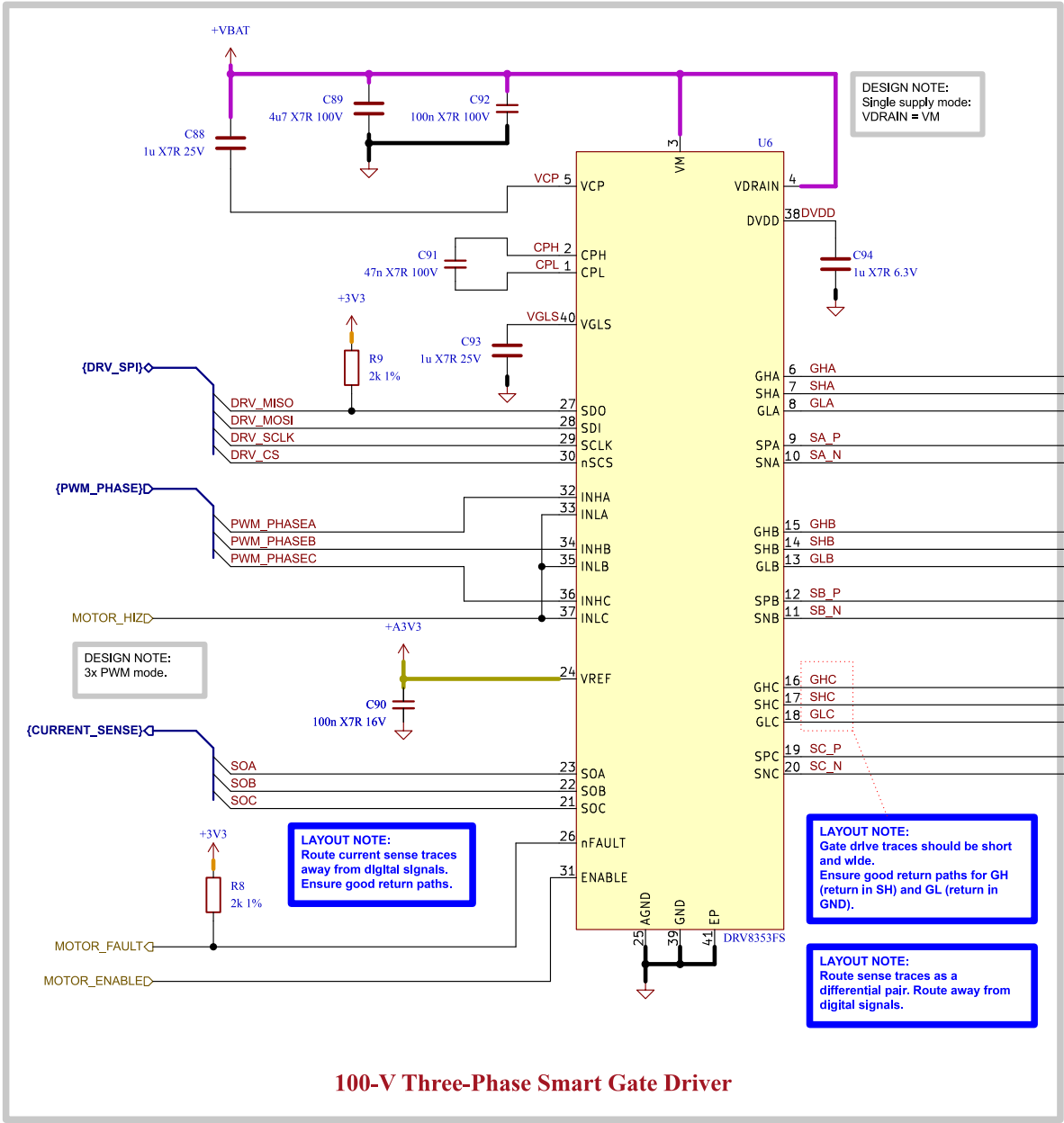
|                                                                                                                                        |                                            |                             |                             |                      |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|-----------------------------|----------------------|
| Comments:<br>+12V (600 mA): Fan power<br>+5V (600 mA): Gate Driver, LEDs<br>+3V3 (500 mA): digital 3.3V<br>+3V3A (250 mA): analog 3.3V | Company:<br>EPFL Xplore Research           |                             | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                                                                                                                                        | Board Name:<br>Amulet Motion Controller    |                             | Project Name:<br>Chienpanzé |                      |
| Sheet Title:<br>Power - Generation                                                                                                     | File Name:<br>Power - Generation.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1     |
| Sheet Path:<br>/Project Architecture/Power - Generation/                                                                               | Reviewer:                                  |                             | Size:<br>A3                 | Sheet:<br>6 of 21    |

# [7] Power - Connectors



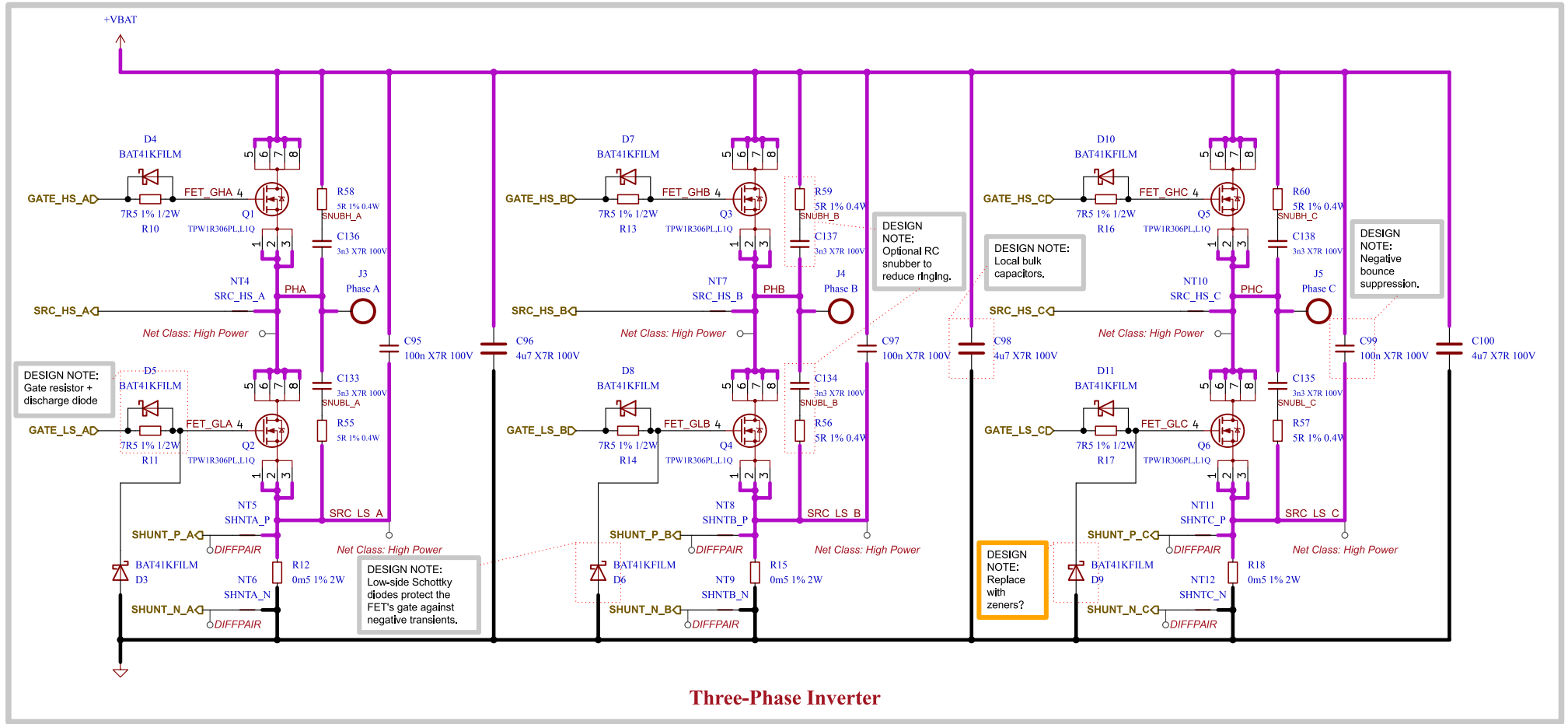
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|  | Comments:                                                | Company:<br>EPFL Xplore Research               |  | Variant:<br>NO_XT60                | Git Hash:<br>74ecd17            |
|  |                                                          | Board Name:<br><b>Amulet Motion Controller</b> |                                                                                       | Project Name:<br><b>Chienpanzé</b> |                                 |
|  | Sheet Title:<br>Power - Connectors                       | File Name:<br>Power - Connectors.kicad_sch     | Designer:<br>Vincent Nguyen                                                           | Date:<br>2023-12-31                | Revision:<br>1.1                |
|  | Sheet Path:<br>/Project Architecture/Power - Connectors/ |                                                | Reviewer:                                                                             | Size:<br><b>A4</b>                 | Sheet:<br><b>7</b> of <b>21</b> |

# [8] Motor Control - Top Level



|                                                  |              |                                     |                          |                |                                                                                       |            |               |         |           |  |
|--------------------------------------------------|--------------|-------------------------------------|--------------------------|----------------|---------------------------------------------------------------------------------------|------------|---------------|---------|-----------|--|
|                                                  | Comments:    |                                     | Company:                 |                |  |            | Variant:      |         | Git Hash: |  |
|                                                  |              |                                     | EPFL Xplore Research     |                |                                                                                       |            | NO_XT60       |         | 74ecd17   |  |
|                                                  |              |                                     | Board Name:              |                |                                                                                       |            | Project Name: |         |           |  |
|                                                  |              |                                     | Amulet Motion Controller |                |                                                                                       |            | Chienpanzé    |         |           |  |
|                                                  | Sheet Title: |                                     | File Name:               |                | Designer:                                                                             |            | Date:         |         | Revision: |  |
| Motor Control - Top Level                        |              | Motor Control - Top Level.kicad_sch |                          | Vincent Nguyen |                                                                                       | 2023-12-20 |               | 1.1     |           |  |
| Sheet Path:                                      |              |                                     |                          | Reviewer:      |                                                                                       | Size:      |               | Sheet:  |           |  |
| /Project Architecture/Motor Control - Top Level/ |              |                                     |                          |                |                                                                                       | A3         |               | 8 of 21 |           |  |

# [9] Motor Control - Inverter



**LAYOUT NOTE:**  
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

**LAYOUT NOTE:**  
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

**DESIGN NOTE:**  
A gate drive current that is too large can damage the FETs!

**Comments:**  
System Design Considerations for High-Power Motor Driver Applications  
Best Practices for Board Layout of Motor Drivers  
Proper RC Snubber Design for Motor Drivers

**Sheet Title:**  
Motor Control - Inverter

**Sheet Path:**  
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

**Company:**  
EPFL Xplore Research



**Board Name:**  
Amulet Motion Controller

**File Name:**  
Motor Control - Inverter.kicad\_sch

**Reviewer:**

**Designer:**  
Vincent Nguyen

**Variant:**  
NO\_XT60

**Project Name:**  
Chienpanzé

**Date:**  
2024-01-25

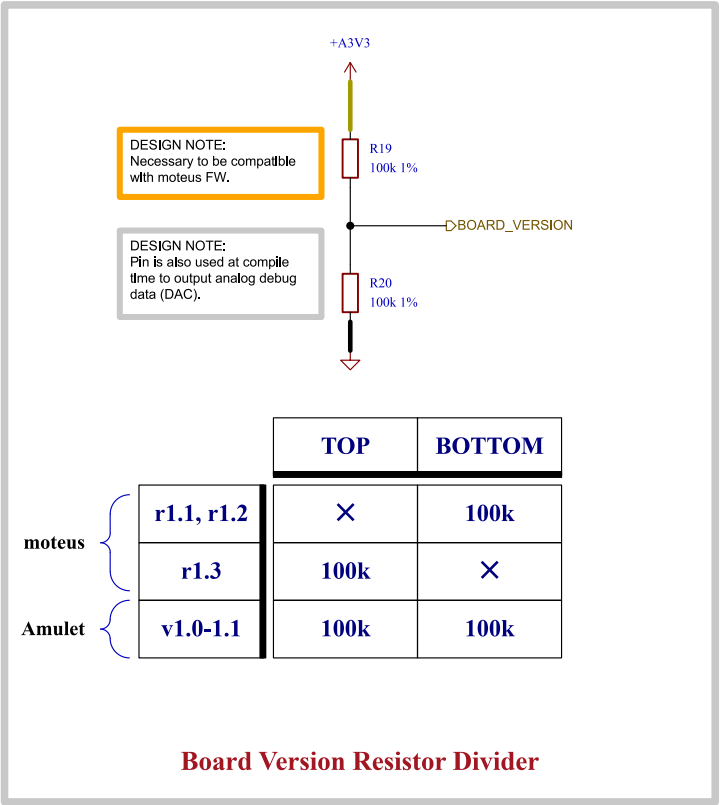
**Size:**  
A4

**Git Hash:**  
74ecd17

**Revision:**  
1.1

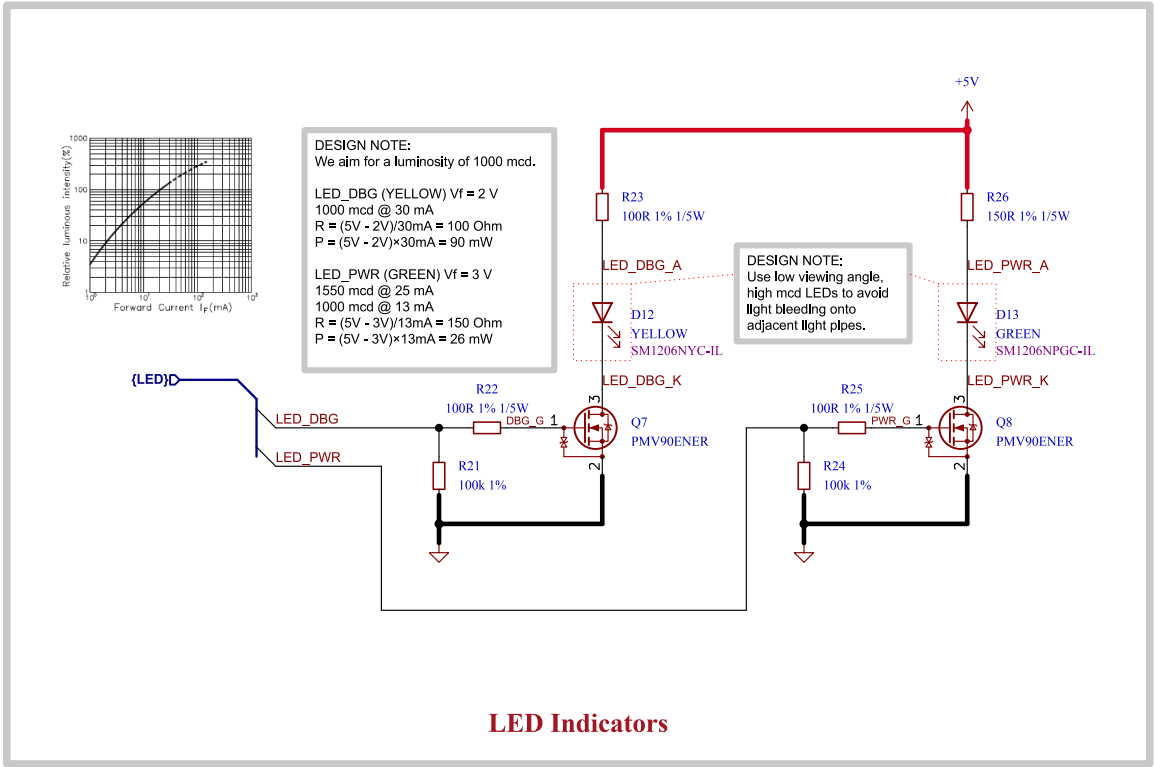
**Sheet:**  
9 of 21

# [10] Misc - Board Version, DAC



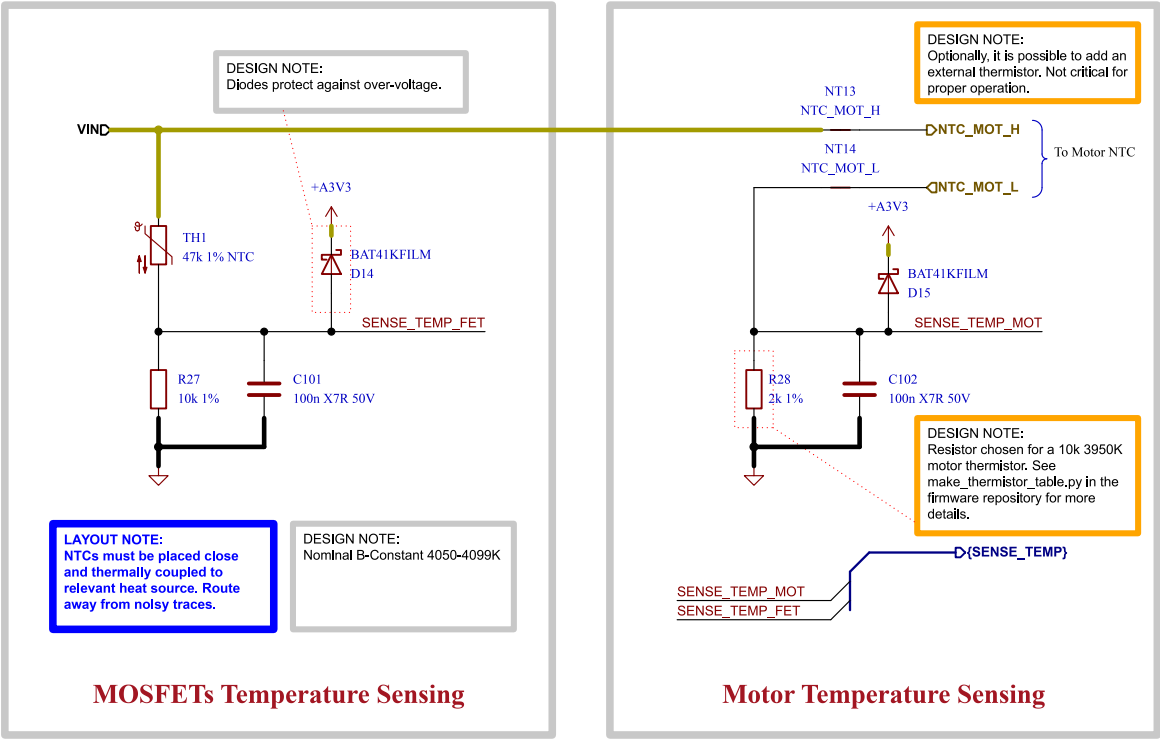
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|--|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------|-------------------------------|
|  | Comments:                                                    | Company: EPFL Xplore Research  |                          | Variant: NO_XT60                | Git Hash: 74ecd17             |
|  |                                                              | Board Name: <b>Amulet Motion Controller</b>                                                                         |                          | Project Name: <b>Chienpanzé</b> |                               |
|  | Sheet Title: Misc - Board Version, DAC                       | File Name: Misc - Board Version DAC.kicad_sch                                                                       | Designer: Vincent Nguyen | Date: 2024-04-13                | Revision: 1.1                 |
|  | Sheet Path: /Project Architecture/Misc - Board Version, DAC/ |                                                                                                                     | Reviewer:                | Size: <b>A4</b>                 | Sheet: <b>10</b> of <b>21</b> |

# [11] User - LED Indicators



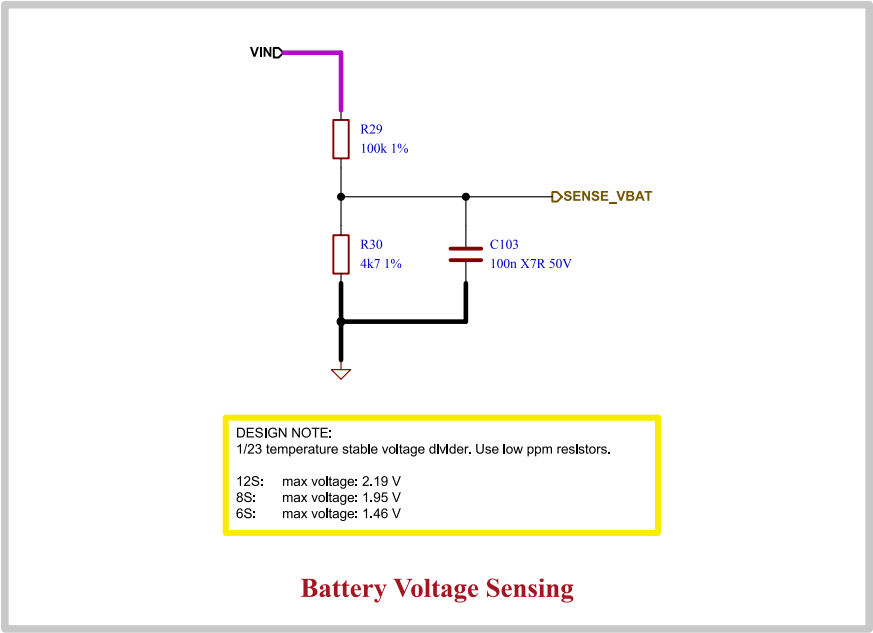
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|  | Comments:                                                   | Company:<br>EPFL Xplore Research              |                             | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             | Project Name:<br>Chienpanzé |                      |
|  | Sheet Title:<br>User - LED Indicators                       | File Name:<br>User - LED Indicators.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-19         | Revision:<br>1.1     |
|  | Sheet Path:<br>/Project Architecture/User - LED Indicators/ |                                               | Reviewer:                   | Size:<br>A4                 | Sheet:<br>11 of 21   |

[12] Sensing - Temperature



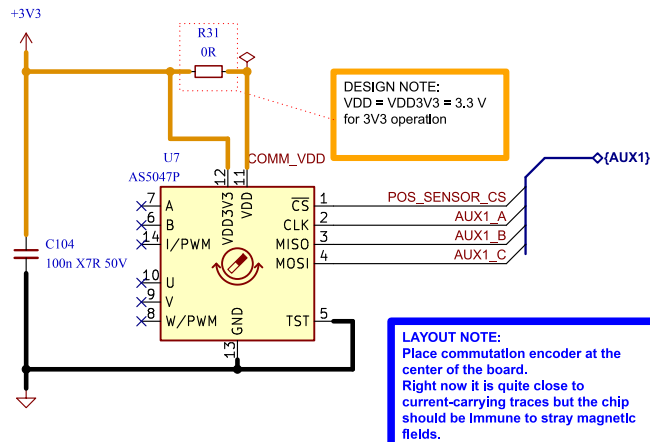
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|  | Comments:                                                   | Company:<br>EPFL Xplore Research              |                             | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|  |                                                             | Board Name:<br>Amulet Motion Controller       |                             | Project Name:<br>Chienpanzé |                      |
|  | Sheet Title:<br>Sensing - Temperature                       | File Name:<br>Sensing - Temperature.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2024-04-13         | Revision:<br>1.1     |
|  | Sheet Path:<br>/Project Architecture/Sensing - Temperature/ |                                               | Reviewer:                   | Size:<br>A4                 | Sheet:<br>12 of 21   |

# [13] Sensing - Battery



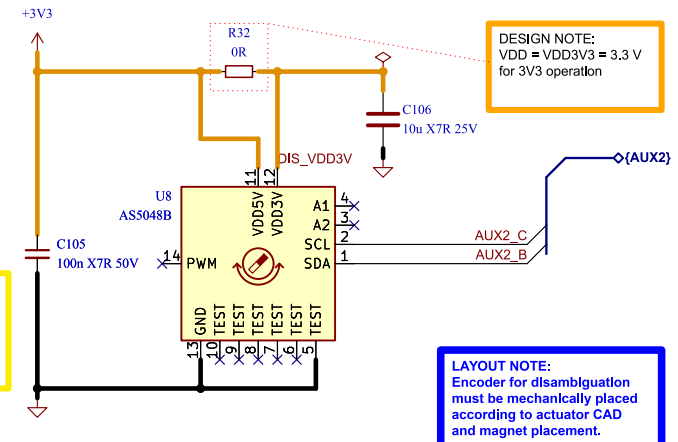
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|  | Comments:                                               | Company:<br>EPFL Xplore Research          |                             |  | Variant:<br>NO_XT60         |                     | Git Hash:<br>74ecd17 |                    |  |
|  |                                                         | Board Name:<br>Amulet Motion Controller   |                             |                                                                                       | Project Name:<br>Chienpanzé |                     |                      |                    |  |
|  | Sheet Title:<br>Sensing - Battery                       | File Name:<br>Sensing - Battery.kicad_sch | Designer:<br>Vincent Nguyen |                                                                                       |                             | Date:<br>2023-10-14 |                      | Revision:<br>1.1   |  |
|  | Sheet Path:<br>/Project Architecture/Sensing - Battery/ |                                           |                             | Reviewer:                                                                             |                             |                     | Size:<br>A4          | Sheet:<br>13 of 21 |  |

# [14] Sensing - Position



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)

DESIGN NOTE:  
AS5047P senses magnet mounted on planetary sun gear, for commutation.  
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.



14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)

|  |                                                          |                                                |                                    |                      |
|--|----------------------------------------------------------|------------------------------------------------|------------------------------------|----------------------|
|  | Comments:                                                | Company:<br>EPFL Xplore Research               | Variant:<br>NO_XT60                | Git Hash:<br>74ecd17 |
|  |                                                          | Board Name:<br><b>Amulet Motion Controller</b> | Project Name:<br><b>Chienpanzé</b> |                      |
|  | Sheet Title:<br>Sensing - Position                       | File Name:<br>Sensing - Position.kicad_sch     | Designer:<br>Vincent Nguyen        | Date:<br>2023-10-14  |
|  | Sheet Path:<br>/Project Architecture/Sensing - Position/ | Reviewer:                                      | Size:<br><b>A4</b>                 | Revision:<br>1.1     |
|  |                                                          |                                                | Sheet:<br><b>14</b> of <b>21</b>   |                      |

[15] Interface - RS-422

**DESIGN NOTE:**  
RS-422 through UART Is used mostly for absolute position sensors (e.g. Aksim-2, SSI protocol)

**DESIGN NOTE:**  
100-ohm termination in Interconnects sheet.

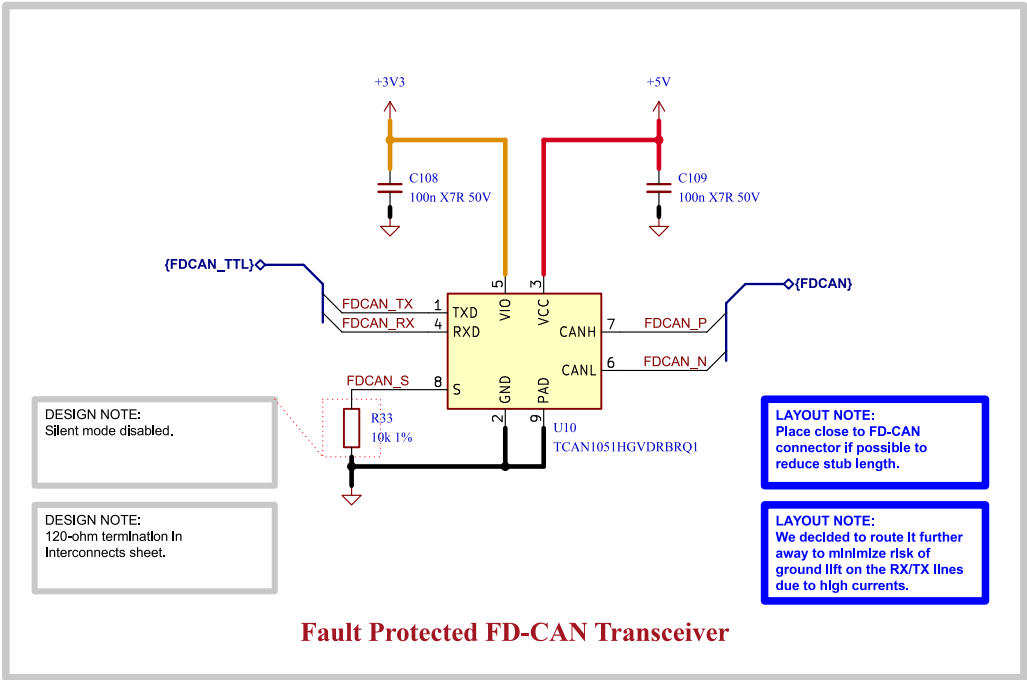
**LAYOUT NOTE:**  
Place close to RS-422 connector.

3.3-V Full-Duplex RS-422 Interface

|  |                                                                 |  |                                                       |  |                                                   |  |                                    |  |                            |  |                                         |
|--|-----------------------------------------------------------------|--|-------------------------------------------------------|--|---------------------------------------------------|--|------------------------------------|--|----------------------------|--|-----------------------------------------|
|  | <b>Comments:</b>                                                |  | <b>Company:</b><br>EPFL Xplore Research               |  | <b>Variant:</b><br>NO_XT60                        |  | <b>Git Hash:</b><br>74ecd17        |  |                            |  |                                         |
|  |                                                                 |  | <b>Board Name:</b><br><b>Amulet Motion Controller</b> |  | <b>Project Name:</b><br><b>Chienpanzé</b>         |  |                                    |  |                            |  |                                         |
|  | <b>Sheet Title:</b><br>Interface - RS-422                       |  |                                                       |  | <b>File Name:</b><br>Interface - RS-422.kicad_sch |  | <b>Designer:</b><br>Vincent Nguyen |  | <b>Date:</b><br>2023-10-15 |  | <b>Revision:</b><br>1.1                 |
|  | <b>Sheet Path:</b><br>/Project Architecture/Interface - RS-422/ |  |                                                       |  |                                                   |  | <b>Reviewer:</b>                   |  | <b>Size:</b><br><b>A4</b>  |  | <b>Sheet:</b><br><b>15</b> of <b>21</b> |

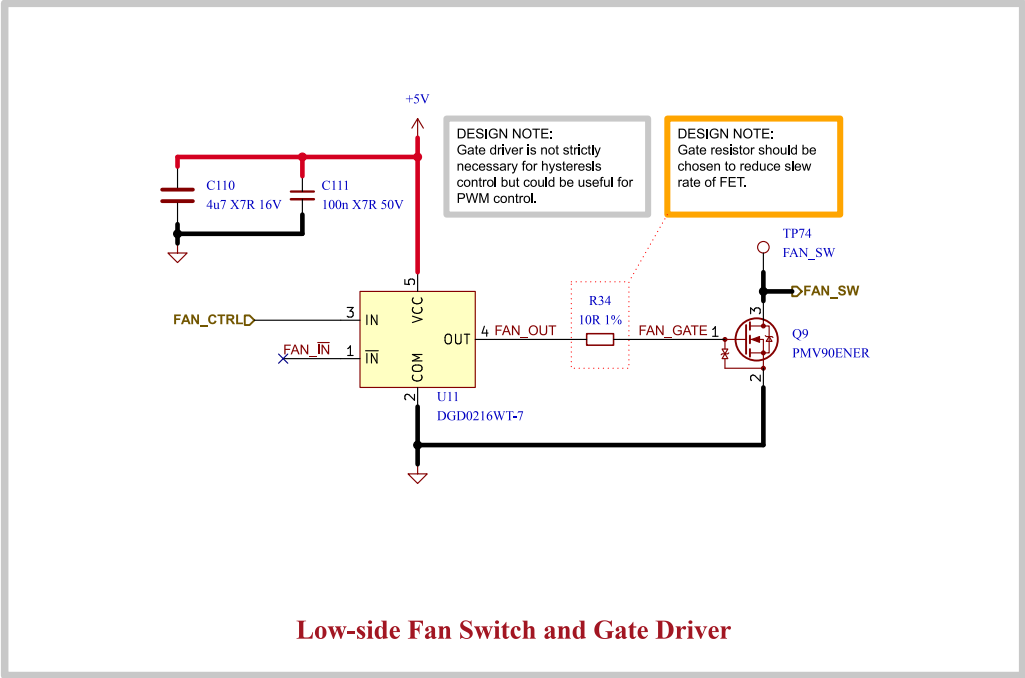
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|                                                          | Comments:                                  | Company:<br>EPFL Xplore Research        |  |  | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                                                          |                                            | Board Name:<br>Amulet Motion Controller |  |                                                                                       | Project Name:<br>Chienpanzé |                      |
| Sheet Title:<br>Interface - RS-422                       | File Name:<br>Interface - RS-422.kicad_sch | Designer:<br>Vincent Nguyen             |  | Date:<br>2023-10-15                                                                   | Revision:<br>1.1            |                      |
| Sheet Path:<br>/Project Architecture/Interface - RS-422/ |                                            | Reviewer:                               |  | Size:<br>A4                                                                           | Sheet:<br>15 of 21          |                      |

# [16] Interface - FD-CAN



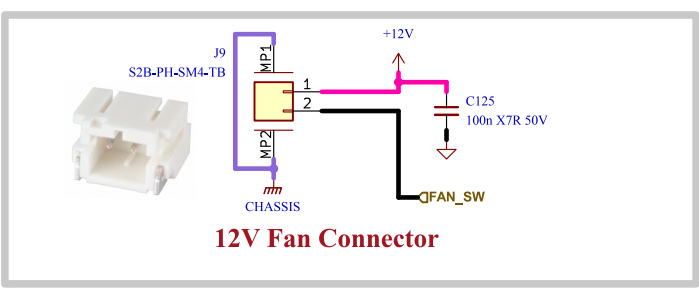
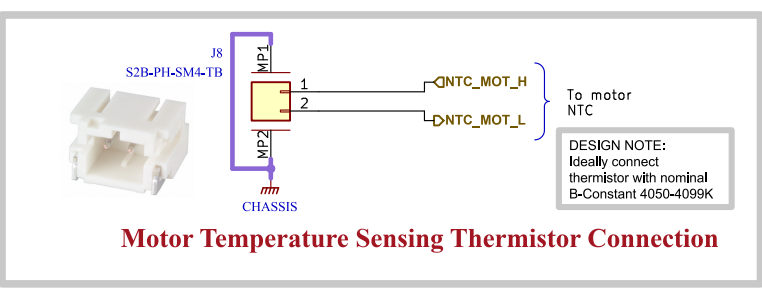
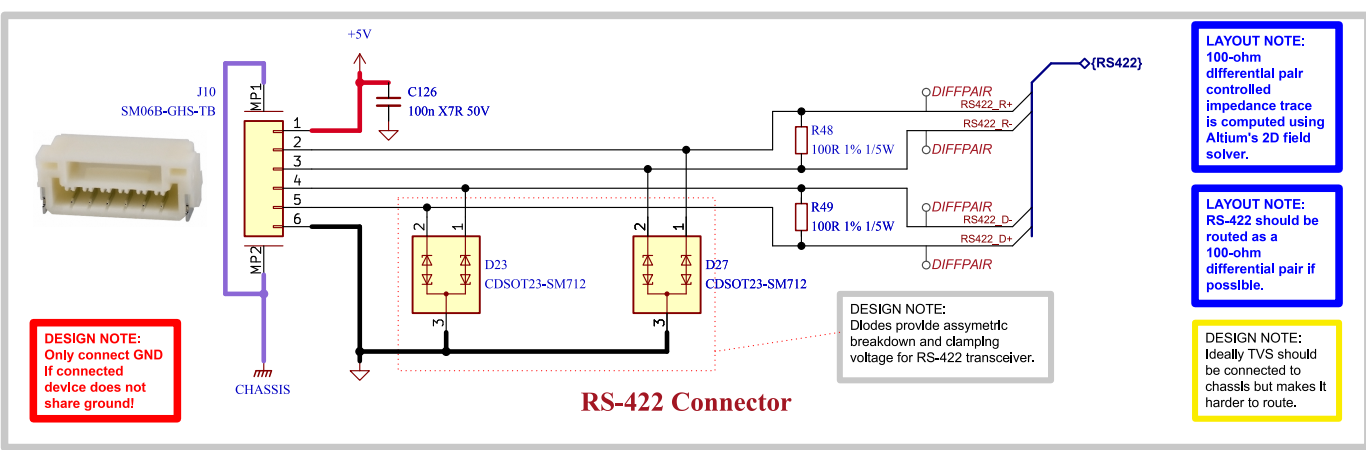
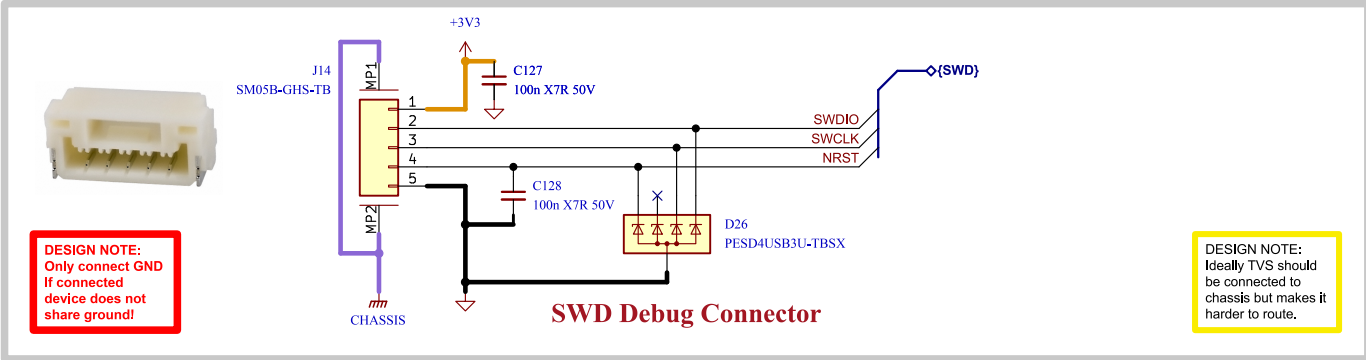
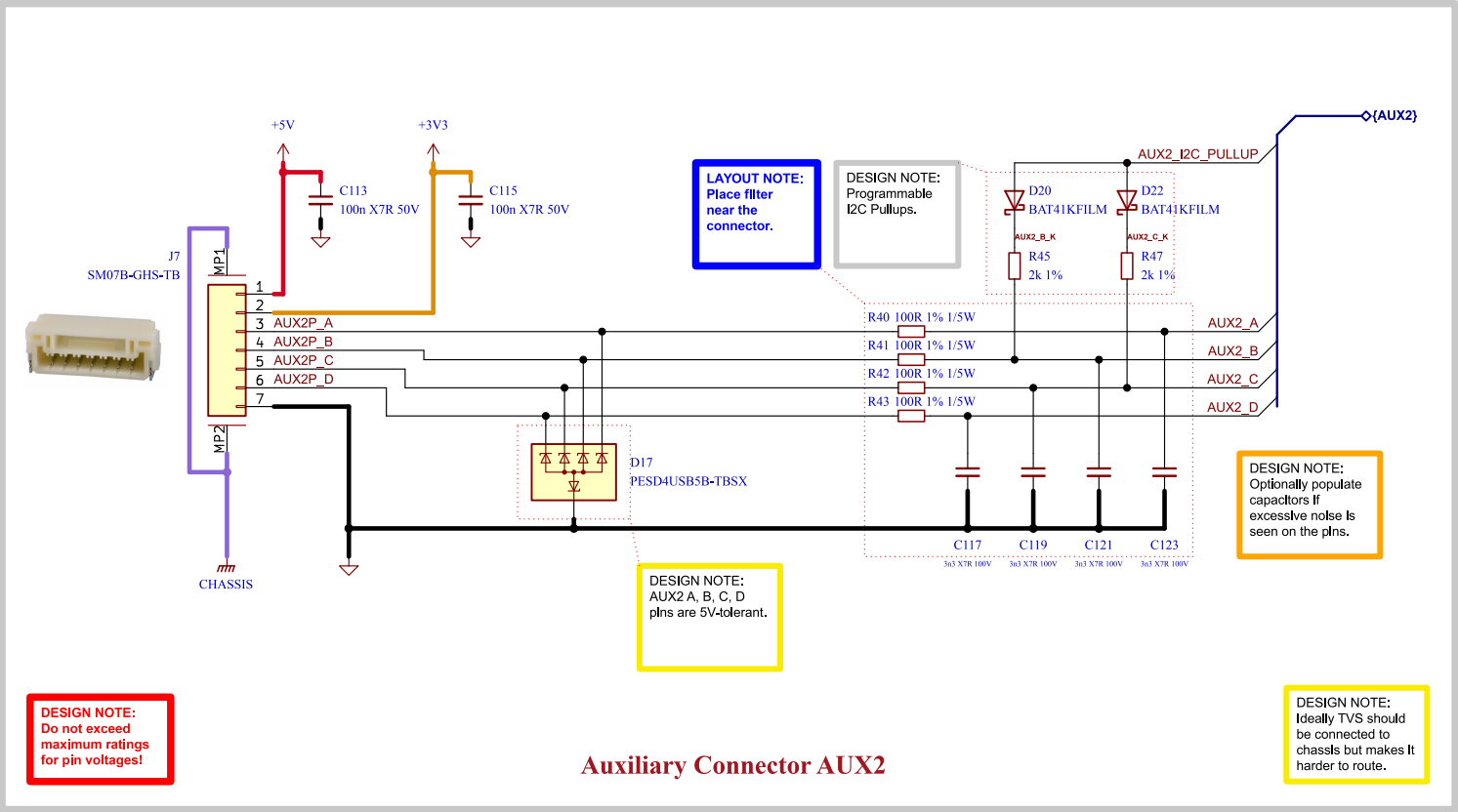
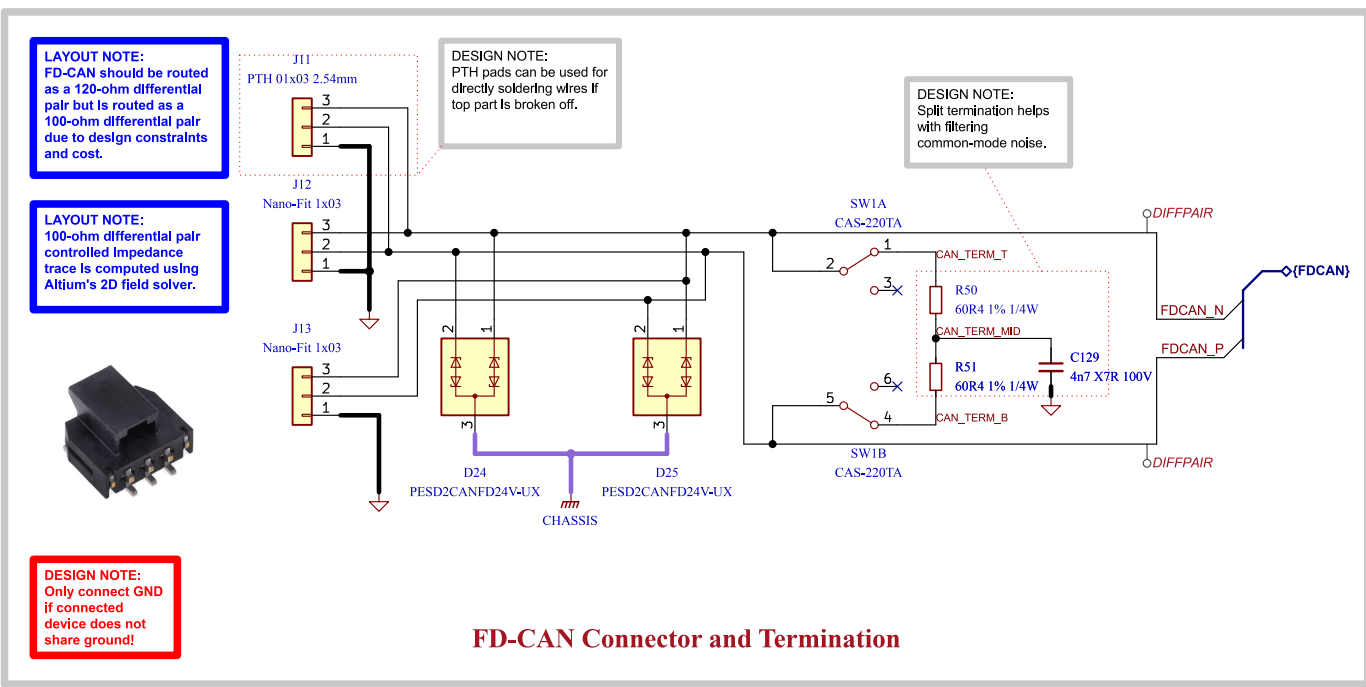
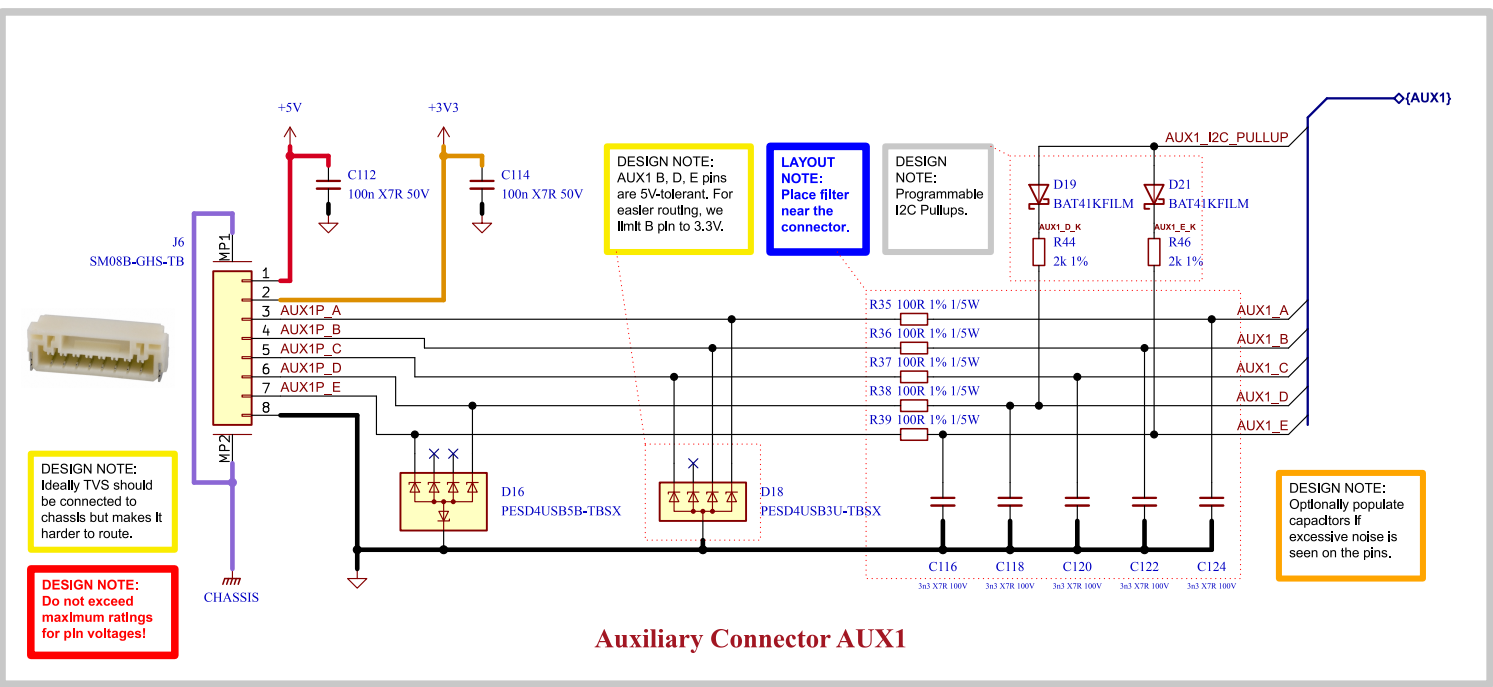
|  |                                                          |                                                |                             |                                    |                                  |
|--|----------------------------------------------------------|------------------------------------------------|-----------------------------|------------------------------------|----------------------------------|
|  | Comments:                                                | Company:<br>EPFL Xplore Research               |                             | Variant:<br>NO_XT60                | Git Hash:<br>74ecd17             |
|  |                                                          | Board Name:<br><b>Amulet Motion Controller</b> |                             | Project Name:<br><b>Chienpanzé</b> |                                  |
|  | Sheet Title:<br>Interface - FD-CAN                       | File Name:<br>Interface - FD-CAN.kicad_sch     | Designer:<br>Vincent Nguyen | Date:<br>2023-10-15                | Revision:<br>1.1                 |
|  | Sheet Path:<br>/Project Architecture/Interface - FD-CAN/ |                                                | Reviewer:                   | Size:<br><b>A4</b>                 | Sheet:<br><b>16</b> of <b>21</b> |

# [17] Interface - Fan Control



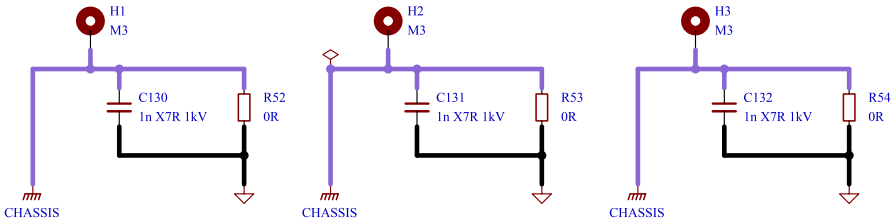
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|--|---------------------------------------------------------------|-------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|-----------------------------|---------------------|----------------------|--------------------|--|
|  | Comments:                                                     | Company:<br>EPFL Xplore Research                |                             |  | Variant:<br>NO_XT60         |                     | Git Hash:<br>74ecd17 |                    |  |
|  |                                                               | Board Name:<br>Amulet Motion Controller         |                             |                                                                                       | Project Name:<br>Chienpanzé |                     |                      |                    |  |
|  | Sheet Title:<br>Interface - Fan Control                       | File Name:<br>Interface - Fan Control.kicad_sch | Designer:<br>Vincent Nguyen |                                                                                       |                             | Date:<br>2023-11-19 |                      | Revision:<br>1.1   |  |
|  | Sheet Path:<br>/Project Architecture/Interface - Fan Control/ |                                                 |                             | Reviewer:                                                                             |                             |                     | Size:<br>A4          | Sheet:<br>17 of 21 |  |

# [18] Interface - Interconnects



|                                                         |                                                                 |                                                   |                             |                      |
|---------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------|-----------------------------|----------------------|
| Comments:<br>Reference:<br>Flexible I/O worked examples | Company:<br>EPFL Xplore Research                                |                                                   | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                                                         | Board Name:<br>Amulet Motion Controller                         |                                                   | Project Name:<br>Chienpanzé |                      |
|                                                         | Sheet Title:<br>Interface - Interconnects                       | File Name:<br>Interface - Interconnects.kicad_sch | Designer:<br>Vincent Nguyen | Date:<br>2023-12-31  |
|                                                         | Sheet Path:<br>/Project Architecture/Interface - Interconnects/ | Reviewer:                                         | Size:<br>A3                 | Revision:<br>1.1     |
|                                                         |                                                                 |                                                   | Sheet:<br>18 of 21          |                      |

# [19] Misc - Holes, Fiducials



DESIGN NOTE:  
Optional RC filter for chassis  
grounding. Capacitors offer multi-point  
low-impedance connections at  
high-frequencies.

DESIGN NOTE:  
0-Ohm resistor provides low-impedance  
bond between chassis and signal GND.  
Be aware of ground loops through cable  
shield.

Mounting Holes and Chassis Connection

Front



Back



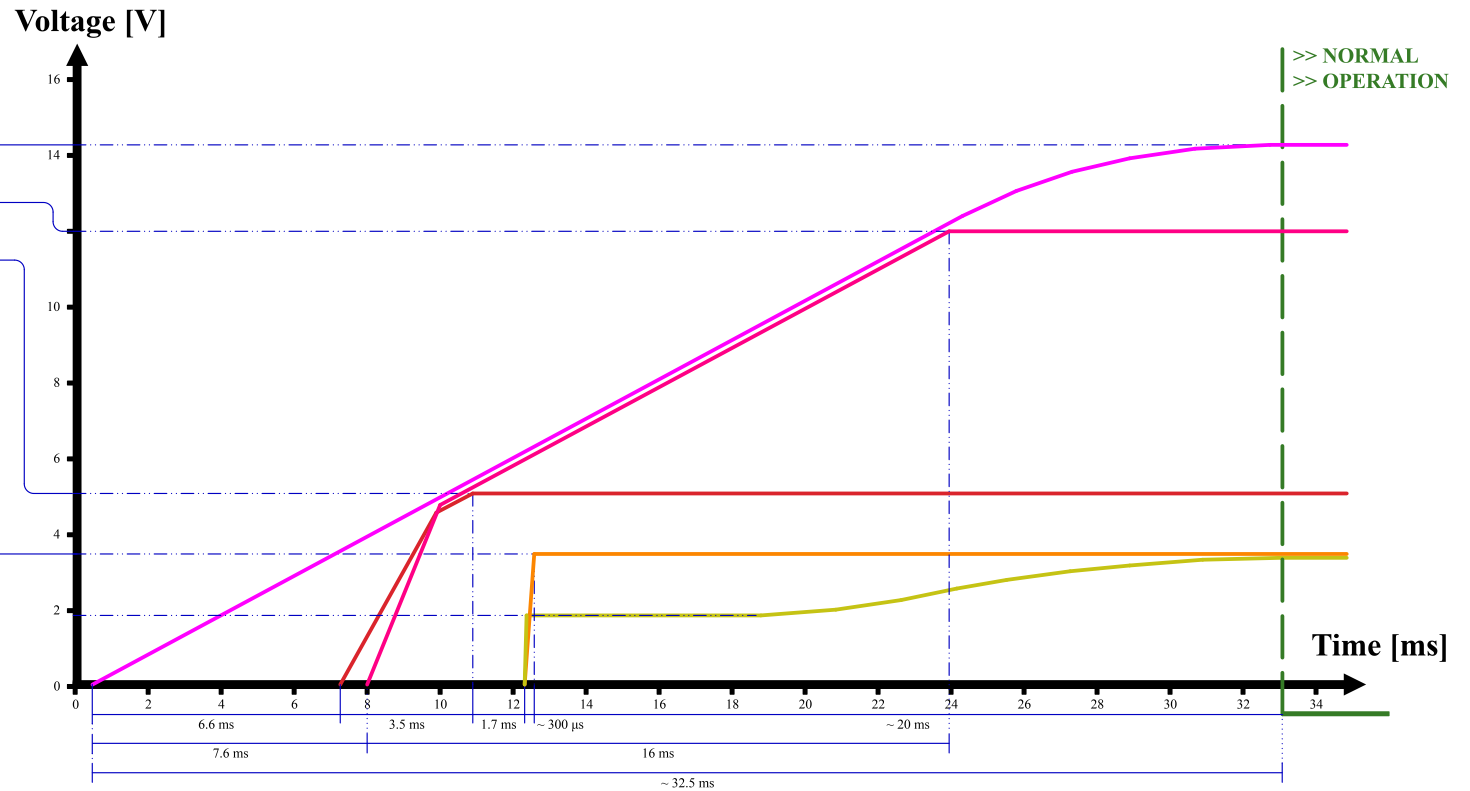
Fiducial Markers

|  |                                                               |                                                |                             |                                                                                       |                     |                             |                      |
|--|---------------------------------------------------------------|------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------|---------------------|-----------------------------|----------------------|
|  | Comments:                                                     | Company:<br>EPFL Xplore Research               |                             |  |                     | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|  |                                                               | Board Name:<br>Amulet Motion Controller        |                             |                                                                                       |                     | Project Name:<br>Chienpanzé |                      |
|  | Sheet Title:<br>Misc - Holes, Fiducials                       | File Name:<br>Misc - Holes Fiducials.kicad_sch | Designer:<br>Vincent Nguyen |                                                                                       | Date:<br>2023-10-22 | Revision:<br>1.1            |                      |
|  | Sheet Path:<br>/Project Architecture/Misc - Holes, Fiducials/ |                                                | Reviewer:                   |                                                                                       | Size:<br>A4         | Sheet:<br>19 of 21          |                      |

[illegible]

**DESIGN NOTE:**  
Power sequencing graph measured experimentally.

**DESIGN NOTE:**  
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



|                                     |                                            |                                         |                     |                                                                                       |                             |                      |
|-------------------------------------|--------------------------------------------|-----------------------------------------|---------------------|---------------------------------------------------------------------------------------|-----------------------------|----------------------|
|                                     | Comments:                                  | Company:<br>EPFL Xplore Research        |                     |  | Variant:<br>NO_XT60         | Git Hash:<br>74ecd17 |
|                                     |                                            | Board Name:<br>Amulet Motion Controller |                     |                                                                                       | Project Name:<br>Chienpanzé |                      |
| Sheet Title:<br>Power - Sequencing  | File Name:<br>Power - Sequencing.kicad_sch | Designer:<br>Vincent Nguyen             | Date:<br>2024-03-12 | Revision:<br>1.1                                                                      |                             |                      |
| Sheet Path:<br>/Power - Sequencing/ |                                            | Reviewer:                               | Size:<br>A4         | Sheet:<br>20 of 21                                                                    |                             |                      |

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| [21] Revision History                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                         |                |  |
| <p><b>12-DEC-2023 - Initial Release</b><br/>Variant: v1.0 Preliminary</p> <ul style="list-style-type: none"> <li>- Changed CPH-CPL capacitor to 47nF (gate driver).</li> <li>- Changed FD-CAN transceiver IC.</li> <li>- Changed FETs for top cooled variant.</li> <li>- Added TVS protection and termination switch to FD-CAN.</li> <li>- Added low-side switched 12V 600mA source for external fan.</li> <li>- Added LDO for analog supply.</li> <li>- Changed input power TVS diode to bidirectional and added one diode per connector.</li> <li>- Moved SOx low-pass filter to MCU section. Should be placed near MCU to avoid noise coupling into ADC lines.</li> <li>- Added second onboard I2C magnetic encoder for disambiguation.</li> <li>- Switched PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.</li> <li>- Changed RS422 pinout on connector.</li> <li>- Added ESD protection to all interfaces.</li> <li>- Added overvoltage protection on thermistor ADC inputs.</li> <li>- Changed buck regulators to optimize for low noise.</li> <li>- Added Pi filters to inputs of buck regulators and MCU analog supply.</li> <li>- Added decoupling caps next to power pins of connectors.</li> </ul> | <p><b>25-JAN-2024 - First Revision</b><br/>Variant: v1.0 Checked</p> <ul style="list-style-type: none"> <li>- Added controller target specifications.</li> <li>- Replaced 5V 300mA buck converter with 600mA version.</li> <li>- Added credits to moteus on cover page.</li> <li>- Added optional RC-Snubber to power stage.</li> <li>- Increased chassis length to go around the board.</li> <li>- CAN and power TVS diodes now go to chassis.</li> <li>- Changed clearance between nets to respect IEC60664-1 where possible.</li> <li>- Rectified comment on precharge.</li> <li>- Changed power TVS diode reference designator from "U" to "D".</li> <li>- Replaced chassis-GND capacitor by 1nF 1kV.</li> </ul> | <p><b>12-MAR-2024 - First Revision</b><br/>Variant: v1.0 Released</p> <ul style="list-style-type: none"> <li>- Modified power sequencing graph according to experimental data.</li> </ul> | <p><b>13-APR-2024 - Second Revision</b><br/>Variant: v1.1 Released</p> <ul style="list-style-type: none"> <li>- Added RC snubber passive values.</li> <li>- Added more vias for VBUS and LMR36006 GND pads.</li> <li>- Changed board version voltage reference from +3V3 to +A3V3.</li> <li>- Changed motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.</li> </ul> |                |  |
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